Attention-Based Dynamic Alignment and Dynamic Distribution Adaptation for Remote Sensing Cross-Domain Scene Classification

Cong Yang, Yanni Dong, *Senior Member, IEEE*, Bo Du, *Senior Member, IEEE*, and Liangpei Zhang, *Fellow, IEEE*

***Abstract*— Due to the lack of high-quality labeled data and poor generalization ability of supervised models in remote sensing scene classification, cross-domain scene classification is proposed to better utilize the existing knowledge to classify the unlabeled data. Since there is a data distribution difference between the training (source domain) and test (target domain) sets, many deep domain adaptation methods have been proposed to reduce such distribution discrepancy. However, existing deep domain adaptation methods usually use the discrepancy metric function to align the marginal distribution and do not consider the effect of each sample in different domains on the network weights. In addition, the existing domain adaptation methods cannot adaptively balance the relative importance of marginal and conditional distributions well. To overcome the above shortcomings, we propose a novel attention-based dynamic alignment and dynamic distribution adaptation (ADA-DDA) method to better align the marginal distribution between different distributions by calculating the dynamic weights of each sample in different domains and dynamically balance the relative importance of marginal and conditional distributions. Moreover, the attention mechanism enables purposeful knowledge transfer so that the extracted features can be highly discriminative. The experimental results demonstrate that our proposed method is superior to the other state-of-the-art deep domain adaptation methods in the comparison and outperforms the second place in accuracy by 5.36%.**

***Index Terms*— Attention mechanism, deep domain adaptation, dynamic distribution alignment, remote sensing, scene classification.**

NOMENCLATURE

$\mathcal{X}_s$, $\mathcal{X}_t$	Feature space of the source domain and the target domain.
$\mathcal{Y}_s$, $\mathcal{Y}_t$	Label space of the source domain and the target domain.
$\mathcal{C}$	Total number of categories.
$\mathcal{D}_s$, $\mathcal{D}_t$	Source domain and target domain.
$\mathcal{P}(\mathcal{X}_s)$, $\mathcal{P}(\mathcal{X}_t)$	Marginal distribution of the source domain and the target domain.
$\mathcal{Q}(\mathcal{Y}_s\|\mathcal{X}_s)$, $\mathcal{Q}(\mathcal{Y}_t\|\mathcal{X}_t)$	Conditional distribution of the source domain and the target domain.
$\mathbf{W}$	Overall dynamic weight matrix.
$\mathbf{W}_s$	Static convolutional kernel matrix.
$\mathbf{W}_d$	Dynamic adjustment matrix.
τ	Dynamic scale matrix.
$\boldsymbol{\omega}_i$	ith static matrix.
Z-pool	Function of concatenating the average pooling and max pooling.
H, W, C	Spatial height, width, and channel of the input data $\mathbf{z}$.
ψ	Convolutional mapping.
$d_{\mathrm{MMD}}(\mathcal{D}_s, \mathcal{D}_t)$	Maximum mean distance between the source domain and the target domain.
$d_{\mathrm{CMMD}}(\mathcal{D}_s, \mathcal{D}_t)$	Conditional maximum mean distance between the source domain and the target domain.

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I. INTRODUCTION

LAST few years, remote sensing scene classification (RSSC) has attracted the attention of many scholars, and it is becoming more and more popular. Advanced machine learning and deep learning methods have been used for RSSC [1], [2], [3], [4], [5], [6], [7]. Different from the pixel-level remote sensing image classification, the RSSC method can obtain the high-level semantic features of the entire image in the remote sensing image, thereby discriminating which scene category the image belongs to. Based on this, RSSC methods are widely studied in many real-world applications, such as urban planning, natural hazards detection, and geospatial object detection [8], [9], [10], [11], [12].

With the continuous iterative advancement of remote sensing image sensors, not only the spatial resolution

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of the images acquired by remote sensing satellites is getting higher and higher but the volume of data has also increased several times. Driven by updates in remote sensing imaging technology, RSSC has been given a well-developed opportunity, especially prompting the widespread use of deep learning in RSSC. The existing supervised deep learning approaches, which are trained on numerous labeled data and tested on the data with the same or similar distribution, have significantly improved the accuracy of scene classification. However, acquiring large amounts of labeled training data can be a time-consuming and expensive task for RSSC. Recently, transfer learning, which can reuse the existing knowledge to improve model performance as much as possible with less or no annotated data, is utilized to alleviate lacking annotated data in the model. Simultaneously, transfer learning algorithms have been rapidly applied to RSSC, which makes remote sensing cross-domain scene classification widely studied. Li et al. [13] proposed a deep convolution neural network based on transfer learning (TL-DeCNN) to solve the lack of high-quality labeled data in RSSC, which just fine-tunes classification model on limited high-spatial resolution remote sensing scene samples in a few epochs. In [14], to solve the problem of expensive data labeling, a transfer learning strategy was applied to remote sensing cross-domain scene classification together with neural architecture search network mobile. However, the two methods mentioned above just transfer the learned model parameters and do not consider the data distribution of source and target domains.

Due to the different distributions of different datasets, the generalization ability of the learned model is poor when applied to a new dataset. The large distribution differences among remote sensing images are caused by variations in sensor type, shooting angle, illumination conditions, imaging conditions, and so on [15]. Hence, to alleviate lacking high-quality annotated data and the weak generalized ability of the model, domain adaptation, which is one of the transfer learning methods, has been proposed to solve these two problems by aligning the distribution of label-rich data (i.e., source domain) and label-sparse data (i.e., target domain). The distribution differences between the source domain and the target domain include marginal distribution discrepancy and conditional distribution discrepancy. Marginal distribution discrepancy refers to different data generation mechanisms, such as differences in image data acquired by sensors under different lighting conditions. Conditional distribution discrepancy refers to the posterior probability distribution difference between data and labels, which is also related to the learning task. Since the domain adaptation approaches assume that the source and target domains have the same label space but different data probability distributions (i.e., domain shift), then the most straightforward way to alleviate the domain shift phenomenon is to transform source and target domain data, which brings the different data distributions closer together. Depending on the nature of the data distribution, such methods can be further classified as marginal distribution adaptation [16], [17], [18], conditional distribution adaptation [19], and joint distribution adaptation [20], [21], [22].

Domain adaptation algorithms, which can align the data distributions of the source and target domains, are applied to remote sensing cross-domain scene classification to weaken the influence of domain shift. Sun et al. [23] utilized a discriminative cross-view subspace alignment algorithm to align the source and target domain data distributions by minimizing a Bregman matrix divergence between the source subspace and the target subspace. Song et al. [24] proposed a subspace alignment layer, which is based on a convolutional neural network, to minimize the discrepancy between source and target domains. Despite the great success of these methods, they only attempt to align the marginal distribution. Thus, Othman et al. [25] designed a domain adaptation network to minimize the marginal distribution discrepancy while proposing a minibatch gradient-based optimization with a dynamic sample size to reduce the conditional distribution discrepancy. Zhang et al. [26] presented a correlation subspace dynamic distribution alignment (CS-DDA) to avoid mapping source domain data into irrelevant subspace while balancing the interclass and intraclass distribution between domains. Although these two methods align both marginal distribution and conditional distribution, they just assign equal weights to the two distribution divergences, which fails to evaluate the relative importance of marginal and conditional distributions. Zhu et al. [27] proposed an attention-based multiscale residual adaptation network (AMRAN) to consider the relative importance of marginal and conditional distributions by utilizing the attention mechanism and the multiscale strategy, which obtained a significant improvement.

However, for remote sensing cross-domain scene classification tasks, there are still three major challenges in existing methods as follows.

1) Although the existing methods can align the marginal distribution discrepancies of the two domains to achieve a better performance, these methods only use the maximum mean discrepancy (MMD) [28] loss function to constrain the marginal distribution discrepancies and do not consider the impact of each sample in the source and target domains. Since each sample in the source and target domains is different, the traditional domain adaptation feature extraction modules, such as ResNet [29] and AlexNet [30], need to be improved to extract domain-invariant features by considering the influence of each sample on the model parameters.
2) The transfer of existing knowledge is often blind. For a training image, it is difficult for the existing feature extraction network to enhance the weight of the key region and weaken the weight of the background value, but this is very useful for obtaining discriminative semantic features.
3) Balancing the relative importance of marginal and conditional distributions is not easy. In the existing research, the conditional distribution will be considered as important as the marginal distribution, but this view has been proven to be flawed by experiments [31]. Although Zhu et al. [27] take marginal and conditional distributions into account, they manually adjust the

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balance parameter between the two distributions, which will lead the balance parameter into a local optimum.

To address the above issues, this article proposes an attention-based dynamic alignment and dynamic distribution adaptation (ADA-DDA) to align the marginal distribution while dynamically adjusting the relative importance of marginal and conditional distributions. There are three contributions in this study, which are summarized as follows.

1) To consider the influence of each sample on the model parameters, the proposed approach utilizes four parallel convolutional layers and the squeeze-and-excitation (SE) module to get the dynamic weights of each sample.
2) To make the alignment of the marginal distribution not blind, we utilize the attention mechanism to guide the source and target domains to align the marginal distribution. Thus, the learned domain-invariant attentive features are more discriminative and transferable.
3) The proposed approach takes a marginal distribution and conditional distribution into consideration and dynamically adjusts the relative importance between the two distributions. By dynamically adjusting the relative importance of marginal and conditional distributions, our proposed ADA-DDA can find the optimal solution between the two distributions.

The remaining contents of this article are organized as follows. Section II presents related works about domain adaptation and attention mechanisms. Section III describes the detail of our proposed method. Section IV presents the experimental data and provides elaborate experimental results to demonstrate the superiority of our proposed method. Section V concludes the research that we have done.

II. Related Work

This section mainly introduces the research of domain adaptation in remote sensing cross-domain scene classification and the attention mechanism.

A. Domain Adaptation

In domain adaptation settings, the whole data in the source domain are labeled, while labeled data in the target domain are unavailable. Recent studies focus on mapping the features from two domains into a common latent space, in which the corresponding feature distributions are close. A common approach in existing studies is to map the original features from source and target domains into the same latent feature space, where the corresponding feature distributions are close. Pan et al. [16] proposed a transfer component analysis that utilizes MMD to measure the divergence across domains while learning some transfer components in a reproducing kernel Hilbert space (RKHS). In domain transfer multiple kernel learning (DTMKL) [32], a multiple-kernel MMD was utilized to measure source and target domain discrepancies. By combining the multiple kernels, the nonlinear ability of a multiple-kernel MMD is stronger than a single-kernel MMD, which significantly improves domain adaptation performance. A manifold embedded distribution alignment (MEDA) [33] was proposed to learn a domain-invariant classifier in the Grassmann manifold by minimizing structural risk while dynamically adjusting the relative importance of marginal and conditional distributions. In the adaptive component embedding (ACE) approach [34], ACE mapped data from two domains into a shared domain-invariant subspace by learning adaptive components. When aligning the marginal and conditional distributions, ACE preserved the geometric properties. Moreover, a domain-invariant classifier was learned by optimizing structural risk minimizing in RKHS.

Recently, due to the development of deep neural networks, deep domain adaptation algorithms have shown a blowout growth [35], [36], [37]. Domain adaptive neural network (DaNN) [38], which was a simple neural network only containing two layers of neurons, added an MMD regularization to measure the distance of the source and target distributions. However, due to the shallow network and limited feature representation ability, the transfer effect was limited. Long et al. [39] proposed a conditional domain adversarial network (CDAN) to effectively align different domains of multimodal distributions through a multilinear conditioning strategy to improve the discriminability and entropy conditioning strategy to guarantee the transferability. Pan et al. [40] proposed a transferable prototype network for domain adaptation such that the prototypes of each class in the source and target domains are close in the embedding space, and the prototypes predict similar score distributions on the source and target data, respectively. In [41], the transferable attention for domain adaptation (TADA) method focused on transferable regions or images since forcefully aligning the untransferable regions may lead to negative transfer. Tang and Jia [42] proposed a discriminative adversarial domain adaptation method for solving the problem that separate design methods for task and domain classifiers are limited in aligning the joint distribution of features and categories across domains. Zhu et al. [43] proposed a deep subdomain adaptation network (DSAN) to align the relevant subdomain distributions within the same category across different domains using a local MMD (LMMD). Through theoretical analysis, Cui et al. [44] found that the discrimination and diversity of category prediction also point to the kernel norm of the batch response matrix. Therefore, they proposed batch nuclear-norm maximization (BNM) to improve the performance of the target domain in domain adaptation. However, these methods either consider only one of the marginal and conditional distributions or manually adjust the relative importance of the two distributions.

For remote sensing cross-domain scene classification, three methods, CS-DDA [26], AMRAN [27], and the two-stage adaptation network (TSAN) [45], consider the relative importance of marginal and conditional distributions to align the source domain and the target domain, but they do not balance the relative importance of marginal and conditional distributions well. Specifically, AMRAN, which is an end-to-end domain adaptation network, mainly contains three modules: the residual adaptation module, the attention module, and the multiscale adaptation module. The AMRAN method utilizes a multiscale adaptation module to measure conditional distribution but only utilizes the residual adaptation

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module to align the marginal distribution and does not dynamically adjust the relative importance of the marginal and conditional distributions. Before the multiscale adaptation module, AMRAN uses the convolutional block attention module (CBAM) to carry out "what and where" to transfer. Our method is an improvement on AMRAN, and the specific differences compared with AMRAN are given as follows.

1) For marginal distribution alignment, the impact of sample-level weights does not consider. It will result in incompletely aligning the marginal distribution discrepancies between the source and target domains. On the contrary, we calculate the dynamic weights of each sample in the source and target domains, obtained from the ADA structure, such that the marginal distributions could be better aligned.
2) The attention mechanism in AMRAN separately calculates spatial attention and channel attention but does not calculate the attention between spatial and channel. Thus, the triplet attention mechanism is embedded in the feature extraction network of our proposed method to obtain information between spatial and channel.
3) For the relative importance of marginal and conditional distributions, the relative importance can only be controlled in AMRAN by manually adjusting the hyperparameter. Although the relative importance of marginal distribution and conditional distribution is not easy to balance [24], our ADA-DDA can dynamically adjust the balance parameters and achieve better results.

B. Attention Mechanism

It is well known that the attention mechanism evolves from the human visual system, which is important for human beings to observe and perceive things. The human visual system tends to focus on the important regions of things, rather than analyzing them all.

Recently, the attention mechanism has been widely studied to promote computer vision tasks [41], [46], [47], [48]. The bottleneck attention module (BAM) [49] and CBAM [50] contain both spatial and channel attention modules. The former connects channels and spatial attention in series, while the latter connects channels and spatial attention in parallel, and CBAM is simpler and has fewer parameters than BAM. By designing an SE block to adaptively recalibrate the channelwise feature responses, the SE network (SENet) [51] can model the interdependence between channels.

In domain adaptation, a few works have applied the attention mechanism to learn domain-invariant attentive features that ensure discriminative and transferable knowledge. Kurmi et al. [52] proposed a certainty-based attention method to estimate which regions of the image can be adapted better by utilizing various Bayesian certainty estimates. In [53], an attentive reweighting strategy was proposed to focus on the domain normalization of components that were more susceptible to domain variability and generated deep features with improved domain invariance and discriminability. To explore how to transfer contextual information across two domains, Yang et al. [54] proposed a self-attention-based cross-attention mechanism to capture the context dependencies between two domains and adapt transferable contexts. For better distribution alignment, Chen et al. [55] proposed a novel multilevel attentive adversarial learning framework, which computed the degree of domain confusion at each level and aggregated the features with the attention mechanism determined by individual domain discriminators. However, different from the existing attention mechanism, the attention mechanism used in our paper constructs the intrinsic relationship across spectral, spatial, and channel dimensions of the input tensor, through rotation operations and residual transformations, and encodes interchannel and spatial information with negligible computational overhead. Simultaneously, the attention mechanism used in this article aims at learning domain-invariant attentive features.

III. Proposed Method

Our proposed ADA-DDA method attempts to utilize an ADA structure and an inception adaptation module (IAM) to learn domain-invariant and class-discriminative feature representations such that the target risk can be minimized based on the marginal and conditional distribution alignments. Simultaneously, the dynamic distribution factor (DDF) is used to balance the relative importance of marginal and conditional distributions. The general framework of our proposed method is shown in Fig. 1.

A. Problem Definition

Aiming at the domain adaptation problem in remote sensing cross-domain scene classification, a source dataset $(\mathcal{X}_s, \mathcal{Y}_s) = \{(\mathbf{x}_s^i, y_s^i)\}_{i=1}^{n_s}$ with n_s samples and a target dataset $(\mathcal{X}_t, \mathcal{Y}_t) = \{(\mathbf{x}_t^i, y_t^i)\}_{i=1}^{n_t}$ with n_t samples can be given. $\mathcal{Y}_s$ is the label space with known label information in the source domain $\mathcal{D}_s$, where $y_s^i \in \{1, 2, \ldots, \mathcal{C}\}$. However, $\mathcal{Y}_t$ represents the label space with unknown label information in the target domain $\mathcal{D}_t$. $\mathcal{D}_s$ and $\mathcal{D}_t$ share the same feature space and label space, that is, $\mathcal{X}_s = \mathcal{X}_t$ and $\mathcal{Y}_s = \mathcal{Y}_t$. The marginal probability distribution and conditional probability distribution are different, that is, $\mathcal{P}(\mathcal{X}_s) \neq \mathcal{P}(\mathcal{Y}_t)$ and $\mathcal{Q}(\mathcal{Y}_s|\mathcal{X}_s) \neq \mathcal{Q}(\mathcal{Y}_t|\mathcal{X}_t)$. The goal of deep adaptation is to design a deep neural network to reduce the discrepancies between the source and target domain distributions by utilizing datasets in the source domain while learning the domain-invariant features. To better describe our proposed method, we define some basic notations used in this article in Nomenclature.

B. Attention-Based Dynamic Alignment

Commonly, domain adaptation algorithms use a typical model to extract features from all samples in the source and target domains. The extracted features are all in a latent space where the marginal distribution is aligned by some discrepancy metrics. Due to the huge difference between the samples from the source and target domains, it is difficult for the model to seek an optimal performance on the two domains [56]. Therefore, in this article, an ADA structure is proposed to change the static dependence of the model weights on the input sample $\mathbf{x} \in \mathcal{X}_s \cup \mathcal{X}_t$

$$\mathbf{W}(\mathbf{x}) = \mathbf{W}_s(\mathbf{x}) + \mathbf{W}_d(\mathbf{x}) \tag{1}$$

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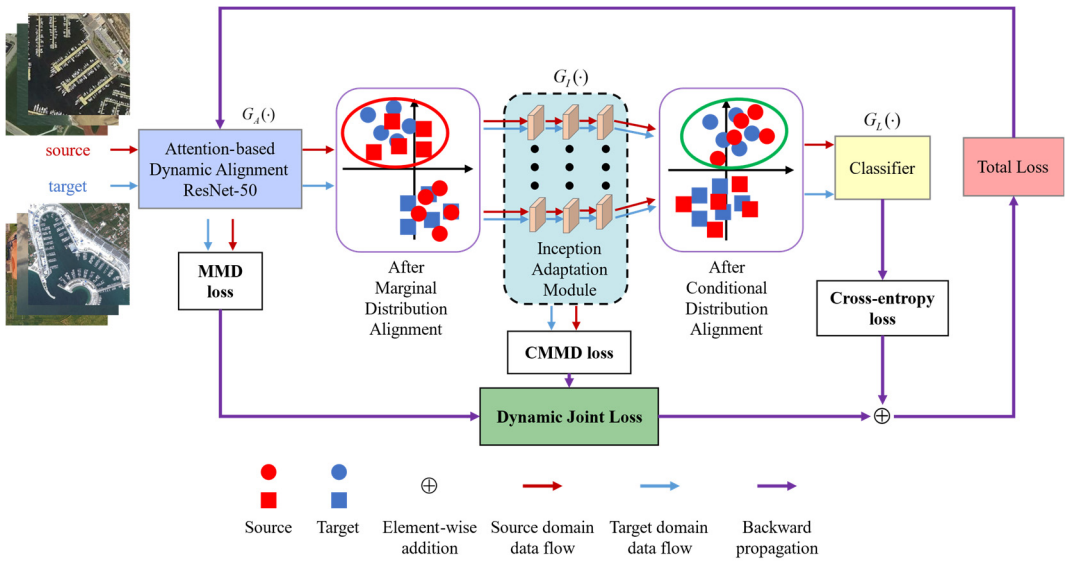


Fig. 1. Overall framework of ADA-DDA. First, the ADA ResNet-50 (50 layers) is utilized to align the marginal distribution divergence while calculating the MMD loss between the source and target domains. Then, an IAM can align the marginal distribution divergence while calculating the CMMD loss between the source and target domains. Simultaneously, the dynamic joint loss is calculated by the MMD loss and the CMMD loss. Finally, the cross-entropy loss of the classifier and the dynamic joint loss are jointly calculated for backward propagation.

where $\mathbf{W}(\mathbf{x})$ represents the overall dynamic weight matrix, $\mathbf{W}_s(\mathbf{x})$ represents the static convolutional kernel matrix, and $\mathbf{W}_d(\mathbf{x})$ is the dynamic adjustment matrix. The specific implementation details are given as follows.

1) Dynamic Alignment Structure: To implement the dynamic alignment structure in the bottleneck of the ResNet, we utilize the altered SE block to emphasize informative features and suppress useless ones for each sample, which is shown in Fig. 2. The altered SE block is composed of adaptative average pooling, two linear layers, and a softmax layer that is to normalize the dynamic scale matrix $\boldsymbol{\tau}$

$$\boldsymbol{\tau} = \varphi(\mathbf{W}_2\delta(\mathbf{W}_1\text{AvgPool}(\mathbf{x}))) \tag{2}$$

where $\mathbf{W}_1$ and $\mathbf{W}_2$ represent the first and second linear layers, respectively. δ refers to the ReLU [57] function, and φ is the softmax function. Then, we use four 1×1 convolutional kernels to obtain the static matrices $\boldsymbol{\omega}_i(\mathbf{x})$, $i \in \{1, \ldots, 4\}$. The dynamic adjustment matrix $\mathbf{W}_d(\mathbf{x})$ can be seen as a linear combination of the dynamic scale matrix and the static matrices in the following equation:

$$\mathbf{W}_d(\mathbf{x}) = \sum_{i=1}^{4} \boldsymbol{\omega}_i(\mathbf{x}) \cdot \boldsymbol{\tau}_i \tag{3}$$

where $\boldsymbol{\tau}_i$ is an element of the dynamic scale matrix τ, and the dynamic adjustment matrix $\mathbf{W}_d(\mathbf{x})$ depends on the input sample $\mathbf{x}$. The static matrices $\boldsymbol{\omega}_i(\mathbf{x})$ can be regarded as the basis of the 1×1 convolution weight space although they are not necessarily linearly independent. The dynamic scale matrix $\boldsymbol{\tau}$ can be seen as projections of the adjustment matrix in the corresponding weight subspaces. By selecting these projections in an input-dependent manner, the network selects different feature subspaces to guide different samples. The combination of the static and dynamic parts forms the dynamic weights of our proposed dynamic alignment structure that can extract the dynamic alignment features. Therefore, (1) can be rewritten as

$$\mathbf{W}(\mathbf{x}) = \mathbf{W}_s(\mathbf{x}) + \sum_{i=1}^{4} \boldsymbol{\omega}_i(\mathbf{x}) \cdot \tau_i. \tag{4}$$

2) Triplet Attention: As shown in Fig. 2, the dynamic alignment feature is fed into the last layer of 1×1 convolution of bottleneck in ResNet, and then, the triplet attention [58] module is used to capture cross-dimension interaction. The triplet attention consists of three parallel branches, two of which separately capture the cross-dimension interaction between the channel dimension and the two spatial dimensions. In addition, the remaining branch is utilized to construct spatial attention, which captures the cross-dimensional interaction between the two spatial dimensions.

The input feature $\mathbf{z} \in \mathbb{R}^{H \times W \times C}$ is obtained from the last layer of 1×1 convolution of the bottleneck. We duplicate it in thirds and input the three replicas into each of the three branches in turn. Before specifically describing the three branches, we introduce the Z-pool layer that preserves the abundant representation of the input feature. The Z-pool layer can reduce the zeroth dimension of the feature to two by concatenating the average pooling function and max pooling function

$$Z\text{-pool}(\mathbf{z}) = \mathcal{Z}(\mathbf{z}) = \left[\text{MaxPool}_{0d}(\mathbf{z}), \text{AvgPool}_{0d}(\mathbf{z})\right] \tag{5}$$

where $0d$ is the zeroth dimension.

In the first branch, to construct the interaction between the height dimension and the channel dimension, the input feature is rotated 90° counterclockwise along the H-axis. The rotated feature is expressed as $\hat{\mathbf{z}}_1$. To obtain the attention weight, the rotated feature passes through the Z-pool layer, a convolutional layer of kernel size 7×7, a batch normalization layer, and a sigmoid activation layer in turn. Subsequently, the attention

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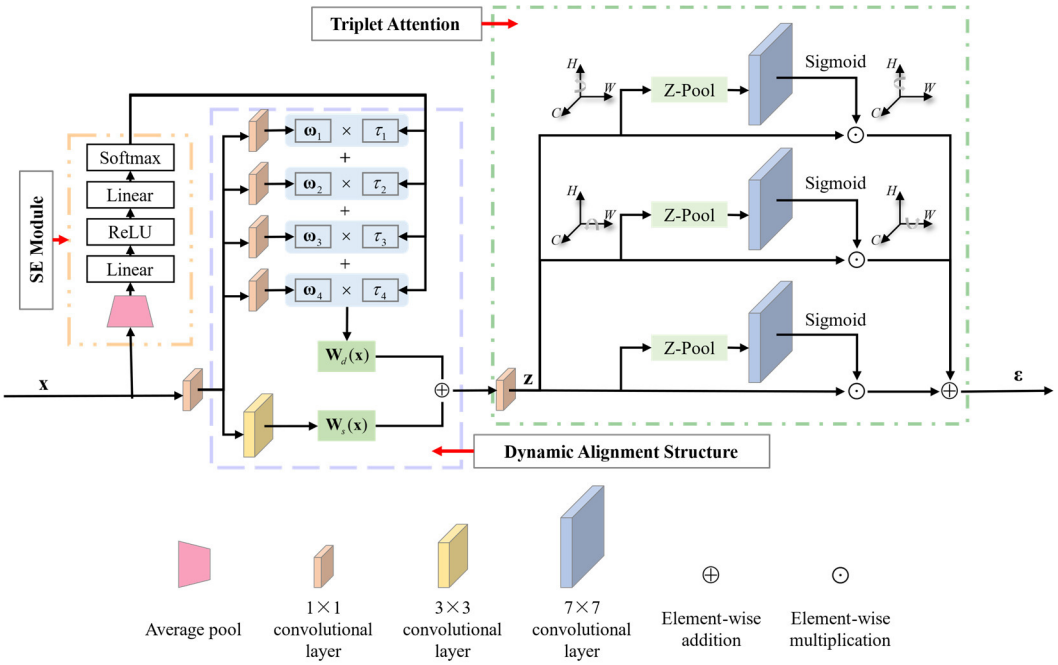


Fig. 2. ADA structure (ADAS). First, ADAS uses the SE module to obtain the four weight values of the sample and calculates the four static weights through four 1×1 convolution blocks at the same time, and then multiplies and adds them to get a dynamic adjustment matrix $\mathbf{W}_d(\mathbf{x})$. Then, elementwise addition of $\mathbf{W}_d(\mathbf{x})$ and $\mathbf{W}_s(\mathbf{x})$ is sent to the triplet attention module to calculate the attention value between the spatial and the channel. Finally, the attention-applied feature ε is sent to the next level building block of ADA Res-Net50.

weight is applied to $\hat{\mathbf{z}}_1$ and then rotated 90° clockwise along the H-axis to restore the original shape of the input feature.

In the second branch, the input feature is rotated 90° counterclockwise along the W-axis. Similarly, the rotated feature $\hat{\mathbf{z}}_2$ is processed similar to the first branch. The input of the third branch is identity mapping $\mathbf{z}$ without any processing. To obtain the attention-applied feature ε, a simple averaging operation is utilized to polymerize the extracted features generated by each of the three branches, which is represented as

$$\boldsymbol{\varepsilon} = \frac{1}{3}\left(\overline{\hat{\mathbf{z}}_1 \sigma\left(\psi_1(\mathcal{Z}(\hat{\mathbf{z}}_1))\right)} + \overline{\hat{\mathbf{z}}_2 \sigma\left(\psi_2(\mathcal{Z}(\hat{\mathbf{z}}_2))\right)} + \mathbf{z}\sigma\left(\psi_3(\mathcal{Z}(\mathbf{z}))\right)\right) \tag{6}$$

where ψ_1, ψ_2, and ψ_3 represent the convolutional layer of kernel size 7×7; the sigmoid function is represented as σ. The 90° clockwise rotation is represented as $\bar{\cdot}$ in (6).

C. Discrepancy Metrics

To align the marginal and conditional distributions, we utilize the MMD and conditional MMD (CMMD) [59], respectively.

1) Maximum Mean Discrepancy: MMD, a nonparametric distribution statistic, is used to calculate the distribution divergence between domains. For the source domain $\mathcal{D}_s$ and the target domain $\mathcal{D}_t$ given above, the MMD loss of the two distributions can be expressed as

$$d_{\mathrm{MMD}}(\mathcal{D}_s, \mathcal{D}_t) = \left\| \frac{1}{n_s} \sum_{\mathbf{x}_s^i \in \mathcal{X}_s} \phi\left(\mathbf{x}_s^i\right) - \frac{1}{n_t} \sum_{\mathbf{x}_t^j \in \mathcal{X}_t} \phi\left(\mathbf{x}_t^j\right) \right\|_{\mathcal{H}_K}^2 \tag{7}$$

where $\mathcal{H}_K$ is the RKHS established by feature mapping $\phi(\cdot)$. The marginal distribution alignment is achieved by minimizing the MMD in (7).

2) Conditional Maximum Mean Discrepancy: Despite reducing the marginal distribution divergence by MMD, the conditional distribution should also be minimized. Thus, a robust domain distribution adaptation needs to minimize the conditional distribution divergence between the source and target domains. Since there is no labeled data in the target domain, directly estimating the conditional distribution $\mathcal{Q}(\mathcal{Y}_t|\mathcal{X}_t)$ of the target domain is difficult. Fortunately, in the deep learning model, the predicted value of the target domain data can be used as the pseudolabel.

Although the posterior probabilities $\mathcal{Q}(\mathcal{Y}_s|\mathcal{X}_s)$ and $\mathcal{Q}(\mathcal{Y}_t|\mathcal{X}_t)$ are difficult to represent, they can be approximated by sufficient statistics of the classwise conditional distributions $\mathcal{Q}(\mathcal{X}_s|\mathcal{Y}_s = c)$ and $\mathcal{Q}(\mathcal{X}_t|\mathcal{Y}_t = c)$. Utilizing the real labels and pseudolabels in the source and target domains, respectively, the MMD can be revised to calculate the distance of the classwise conditional distribution, called CMMD

$$d_{\mathrm{CMMD}}(\mathcal{D}_s, \mathcal{D}_t) = \frac{1}{\mathcal{C}} \sum_{c=1}^{\mathcal{C}} \left\| \frac{1}{n_s^{(c)}} \sum_{\mathbf{x}_{s_{(c)}}^i \in \mathcal{X}_s^{(c)}} \phi\left(\mathbf{x}_{s_{(c)}}^i\right) - \frac{1}{n_t^{(c)}} \sum_{\mathbf{x}_{t_{(c)}}^j \in \mathcal{X}_t^{(c)}} \phi\left(\mathbf{x}_{t_{(c)}}^j\right) \right\|_{\mathcal{H}_K}^2 \tag{8}$$

where $\mathcal{X}_{s_{(c)}}$ represents that the samples belong to the class c in the source domain; likewise, $\mathcal{X}_{t_{(c)}}$ represents that the samples belong to the class c in the target domain.

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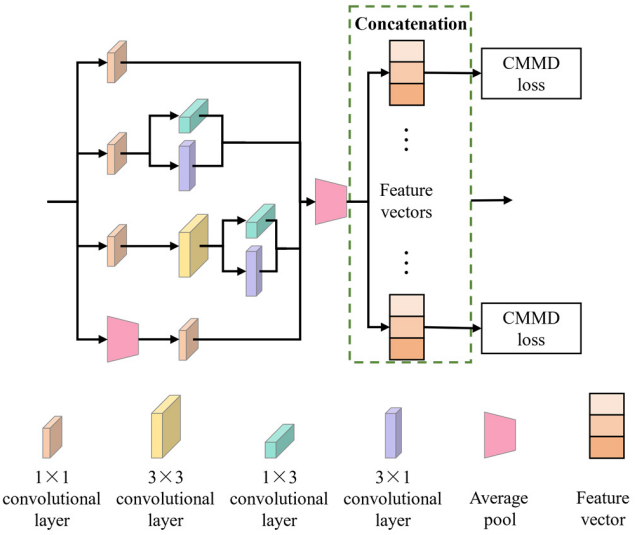


Fig. 3. IAM for multiscale feature extraction and conditional distribution alignment. The four branches, respectively, extract the features of the four levels and calculate the CMMD loss. The adaptation loss of the conditional distribution is the sum of the four substructures. The output features can be obtained through multiscale concatenation. Finally, the multiscale concatenation features are sent to the classifier for better classification.

Moreover, the CMMD is integrated into each substructure of the IAM [59] to minimize the conditional distributions between the source and target domains, which is specifically described in Fig. 3.

D. Dynamic Distribution Factor α

Up until now, we can calculate marginal probability distribution and conditional marginal probability distribution. However, how to dynamically adjust the weight between the two distributions is a crucial issue. Random guessing and average search are the two most ways of acquiring α [31]. By selecting a random α in $[0, 1]$, random guessing utilizes the corresponding value to execute joint distribution to obtain the result, which is repeated several times. The final result is obtained by averaging all the results. An average search uses the average of eleven results that are obtained by selecting the value of $\alpha = 0, 0.1, \ldots, 1.0$ to execute joint distribution eleven times as the final result. However, the computational overhead of these two methods is very large.

In this article, we proposed the DDF to calculate the relative importance of marginal and conditional distributions dynamically and robustly. Specifically, the MMD distance and the CMMD distance can be seen as the marginal and conditional distributions, respectively. Thus, DDF α can be represented as

$$\alpha = \frac{d_{\text{MMD}}(\mathcal{D}_s, \mathcal{D}_t)}{d_{\text{MMD}}(\mathcal{D}_s, \mathcal{D}_t) + \sum_{i=1}^{r} d_{\text{CMMD}}(\mathcal{D}_s, \mathcal{D}_t)} \tag{9}$$

where r represents the total number of branches in IAM. After each epoch of iteration, the DDF will be updated. When training converges, ADA-DDA will learn a robust DDF.

E. Overall Objective Function

The overall structure of ADA-DDA mainly consists of three components: ADA structure ResNet-50 $G_A(\cdot)$, IAM $G_I(\cdot)$, and label classifier $G_L(\cdot)$, as shown in Fig. 1. The marginal distribution adaptation loss $\mathcal{L}_{\text{MD}}$ and conditional distribution adaptation loss $\mathcal{L}_{\text{CD}}$ are calculated as

$$\mathcal{L}_{\text{MD}} = d_{\text{MMD}}(G_A(\mathcal{X}_s), G_A(\mathcal{X}_t)) \tag{10}$$

$$\mathcal{L}_{\text{CD}} = \sum_{i=1}^{r} d_{\text{CMMD}}\left(G_I^i(G_A(\mathcal{X}_s)), G_I^i(G_A(\mathcal{X}_t))\right). \tag{11}$$

The label classifier $G_L(\cdot)$ can discriminate which category the input sample from the source domain belongs to. The cross-entropy loss $\text{CE}(\cdot, \cdot)$ is utilized to train the label classifier, which can be represented as

$$\mathcal{L}_{\text{cls}} = \frac{1}{n_s} \sum_{\mathbf{x}_s^i, y_s^i \in \mathcal{D}_s} \text{CE}\left(G_L\left(G_I\left(G_A\left(\mathbf{x}_s^i\right)\right)\right), y_s^i\right). \tag{12}$$

Integrating all loss functions (10)–(12), the overall objective function can be represented as

$$\mathcal{L}(\theta_A, \theta_I, \theta_L) = \mathcal{L}_{\text{cls}} + \lambda(\alpha\mathcal{L}_{\text{MD}} + (1-\alpha)\mathcal{L}_{\text{CD}}) \tag{13}$$

where λ is a nonnegative tradeoff parameter; θ_A, θ_I, and θ_L are the parameters to be learned for each module. All the parameters to be learned can be denoted as $(\theta_A, \theta_I, \theta_L)$

$$\frac{\Delta\mathcal{L}(\theta_A, \theta_I, \theta_L)}{\Delta(\theta_A, \theta_I, \theta_L)} = \frac{\Delta\mathcal{L}_{\text{cls}}}{\Delta(\theta_A, \theta_I, \theta_L)} - \lambda\frac{\Delta(\alpha\mathcal{L}_{\text{MD}} + (1-\alpha)\mathcal{L}_{\text{CD}})}{\Delta(\theta_A, \theta_I, \theta_L)}. \tag{14}$$

The Adam algorithm [60] can effectively train our proposed method.

IV. Experiments

In this section, we evaluate the proposed ADA-DDA competitive state-of-the-art deep domain adaptation methods. The code of ADA-DDA will be available at https://sites.google.com/view/dongyanni/Home/codes-and-datasets.

A. Cross-Domain Scene Dataset

Since the benchmark data set for cross-domain scene classification is unavailable, we compose the existing three scene classification data sets into a cross-domain scene classification data set, which is described in detail as follows.

1) UC-Merced Dataset: Containing 21 land-use scene classes, the UCM-Merced (UCM) dataset [61] consists of aerial orthoimages with a pixel resolution of 0.3 m, which is available in 2010. Each category consists of 100 images with 256 × 256 pixels. Fig. 4(a) lists eight categories of UCM: *baseball diamond, beach, dense residential, forest, harbor, parking lot, river,* and *storage tanks*. These images are downloaded from the United States Geological Survey National Map of 20 U.S. regions. The categories in UCM contain a variety of spatial patterns, such as homogeneous texture, homogeneous color, or all heterogeneity, which poses a challenge to existing research.

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Fig. 4. Display of each category of sample in each data set. (a) Samples in UCM. (b) Samples in AID. (c) Samples in NWPU. The categories selected from each dataset are *baseball diamond, beach, dense residential, forest, harbor, parking lot, river,* and *storage tanks*.

2) Aerial Image Dataset: By collecting images from Google Earth imagery, Xia et al. [62] construct the large-scale Aerial Image Dataset (AID), which is available in 2017. It consists of 30 aerial scene categories. Fig. 4(b) lists eight categories of AID: *baseball field, beach, dense residential, forest, port, parking, river,* and *storage tanks*. There are approximately 220–420 images with 600 × 600 pixels in each category. Furthermore, the pixel resolution of each scene category varies from about 8 m to about 0.2 m. The AID has higher intraclass variations and smaller interclass dissimilarities that make the AID very challenging.

3) NWPU-RESISC45 Dataset: The NWPU-RESISC45 (NWPU) dataset [63] is available in 2017. The image source of the dataset is the same as AID, which is also collected from Google Earth. It contains 45 scene categories. Fig. 4(c) lists eight categories of NWPU: *baseball diamond, beach, dense residential, forest, harbor, parking lot, river,* and *storage tank*. Each category includes 700 images with 256 × 256 pixels. The spatial resolution varies from about 30 to 0.2 m. The NWPU dataset has three significant characteristics: large scale, rich image variations, high intraclass diversity, and high interclass similarity.

TABLE I
NUMBER OF SAMPLES IN EACH CATEGORY OF EACH DATASET

Class	UCM	AID	NWPU
baseball diamond	100	220	700
beach	100	400	700
dense residential	100	410	700
forest	100	250	700
harbor	100	380	700
parking lot	100	390	700
river	100	410	700
storage tanks	100	360	700
total	800	2820	5600

Since there is no cross-domain scene benchmark dataset, the similar eight categories, renamed to *baseball diamond, beach, dense residential, forest, harbor, parking lot, river,* and *storage tanks*, are selected from each dataset mentioned above. The specific information is shown in Table I.

B. Experimental Setting

The UCM dataset, the AID dataset, and the NWPU dataset are abbreviated as domain U, domain A, and domain N, respectively. Thus, we construct six cross-domain scene classification tasks in the whole experiment denoted by U→A, U→N, A→U, A→N, N→A, and N→U. The arrow "→" represents the direction of knowledge transfer from the source domain to the target domain. DAAN (2019), DeepCORAL (2016) [64], BNM (2020), CDAN (2018), CDAN+E (2018), AMRAN (2021), DSAN (2020), and DeepMEDA (2020) [65] are our comparison algorithms. Our evaluation criterion is overall accuracy and average accuracy (AVG). The average accuracy is the value obtained by averaging the overall accuracy of the six tasks.

We embed the ADA structure into the bottleneck of ResNet-50 (50 layers) and utilize the triplet attention structure in the basic block of ResNet-50. For the comparison algorithms, ResNet50 is directly used to extract deep transferable features. As for the IAM, we directly use the structure in AMRAN without any changes. The detailed structure is substructure1 (conv1 × 1), substructure2 [conv1 × 1 and concat(conv1 × 3, conv3 × 1)], substructure3 [conv1 × 1, conv3 × 3, and concat(conv1 × 3, conv3 × 1)], and substructure4 (AvgPool and conv1 × 1).

Following the standard process, all labeled source data and unlabeled target data are utilized for cross-domain scene classification. All the methods are implemented by the PyTorch framework and trained from scratch with two Nvidia Tesla V100 GPUs. As mentioned above, the Adam algorithm is used for optimization tasks with the momentum setting as 0.9.

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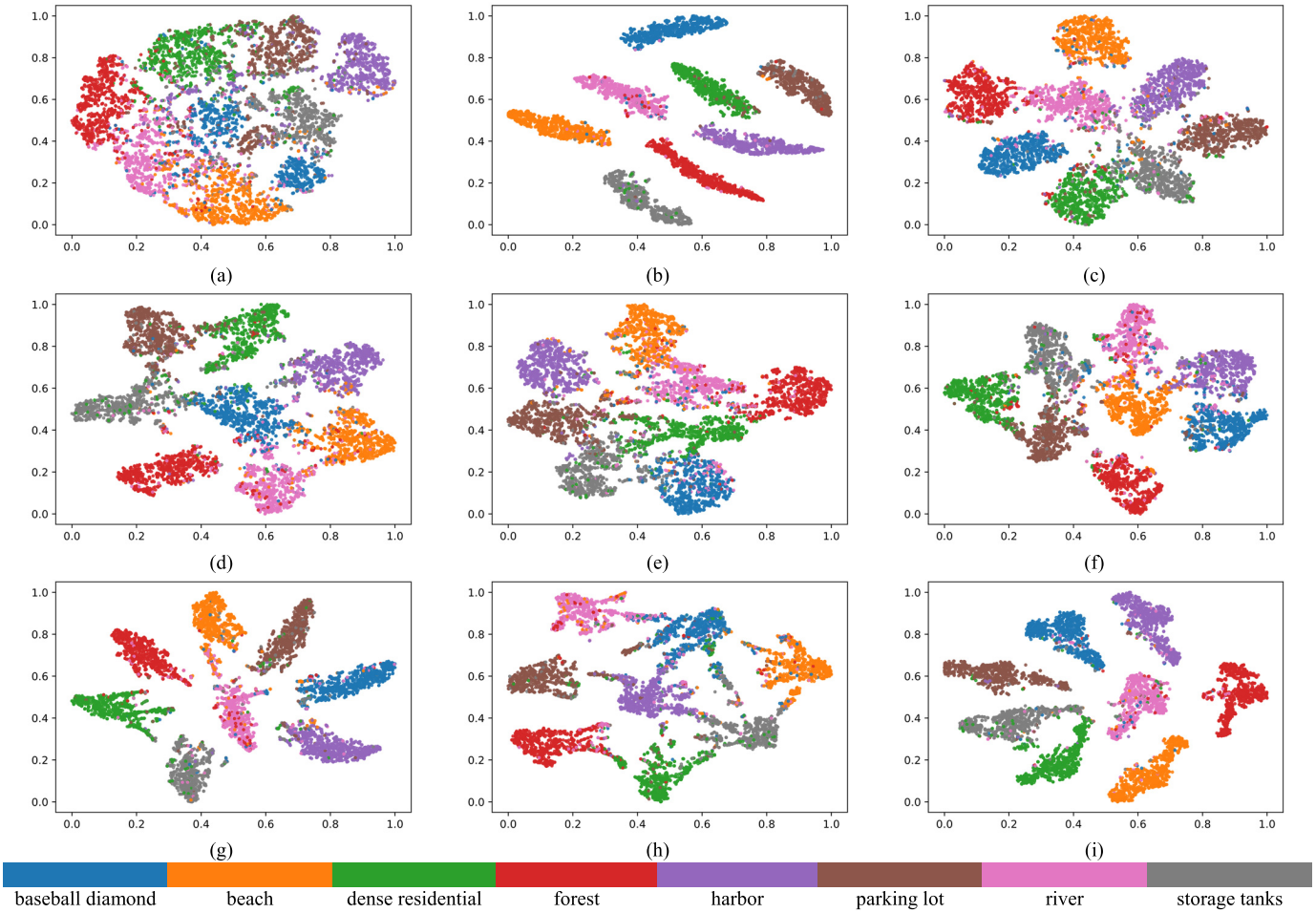


Fig. 5. Visualization: visualized results of t-SNE dimensionality reduction on the source and target domains on the N→U task. (a) DAAN. (b) DeepCORAL. (c) BNM. (d) CDAN. (e) AMRAN. (f) CDAN+E. (g) DSAN. (h) DeepMEDA. (i) ADA-DDA.

Referring to RevGrad, the learning rate annealing strategy is used to stabilize training: it requires a lot of calculations to determine the learning rate through grid search. We can adjust the learning rate during Adam using the following formula: $\ell_p = 0.002/(1+10p)^{0.75}$, where p (p = epoch/epochs) is the training progress that linearly varies from 0 to 1. Simultaneously, the nonnegative tradeoff parameter λ is set according to the formula $\lambda_p = 2/(1+\exp(-10p)) - 1$ in [66]. Since the DDF α is self-adaptive, we do not make any settings. For a fair comparison, all other parameters utilized in all tasks are fixed. Specifically, the training epochs, the batch size, and L2decay in Adam are set as 500, 64, and 0.0005, respectively.

TABLE II

ACCURACY (%) ON EACH CROSS-DOMAIN SCENE CLASSIFICATION TASK FOR DOMAIN ADAPTATION METHODS (RESNET-50 AS BACKBONE)

	U→A	U→N	A→U	A→N	N→U	N→A	AVG
DAAN	49.71	48.50	66.50	77.73	59.88	79.26	63.60
DeepCORAL	74.61	66.50	76.50	84.50	79.38	86.91	78.07
BNM	71.13	69.13	79.50	88.63	71.75	89.43	78.26
CDAN	67.80	66.59	77.63	90.32	76.75	**93.16**	78.71
AMRAN	74.08	68.09	75.50	86.80	78.50	89.43	78.73
CDAN+E	72.05	68.13	73.88	90.32	79.00	92.23	79.27
DSAN	74.65	74.86	73.88	87.05	78.25	88.83	79.58
DeepMEDA	75.18	**75.84**	73.75	89.70	76.63	89.08	80.03
ADA-DDA (ours)	**77.78**	74.76	**87.50**	**90.70**	**89.63**	91.98	**85.39**

C. Comparison and Analysis

1) Comparisons: The experimental results of ADA-DDA and the other compared algorithms are shown in Table II. The proposed ADA-DDA outperforms all the other comparison algorithms while significantly improving the classification accuracy. The detailed analysis is shown in the following.

Except for the U→N and N→A tasks, ADA-DDA outperforms the state-of-the-art approaches on all tasks. On the U→N task, the ADA-DDA still achieves the second-best result. In terms of the AVG, ADA-DDA achieves the best average accuracy of 85.39%, which significantly improves the accuracy. Our proposed method is 21.79%, 7.32%, 7.13%, 6.68%, 6.66%, 6.12%, 5.81%, and 5.36% higher than DAAN, DeepCORAL, BNM, CDAN, AMRAN, CDAN+E, DSAN, and DeepMEDA, respectively. Compared with AMRAN, when there are more categories in the source and target domains, the performance of ADA-DDA is much better than AMRAN.

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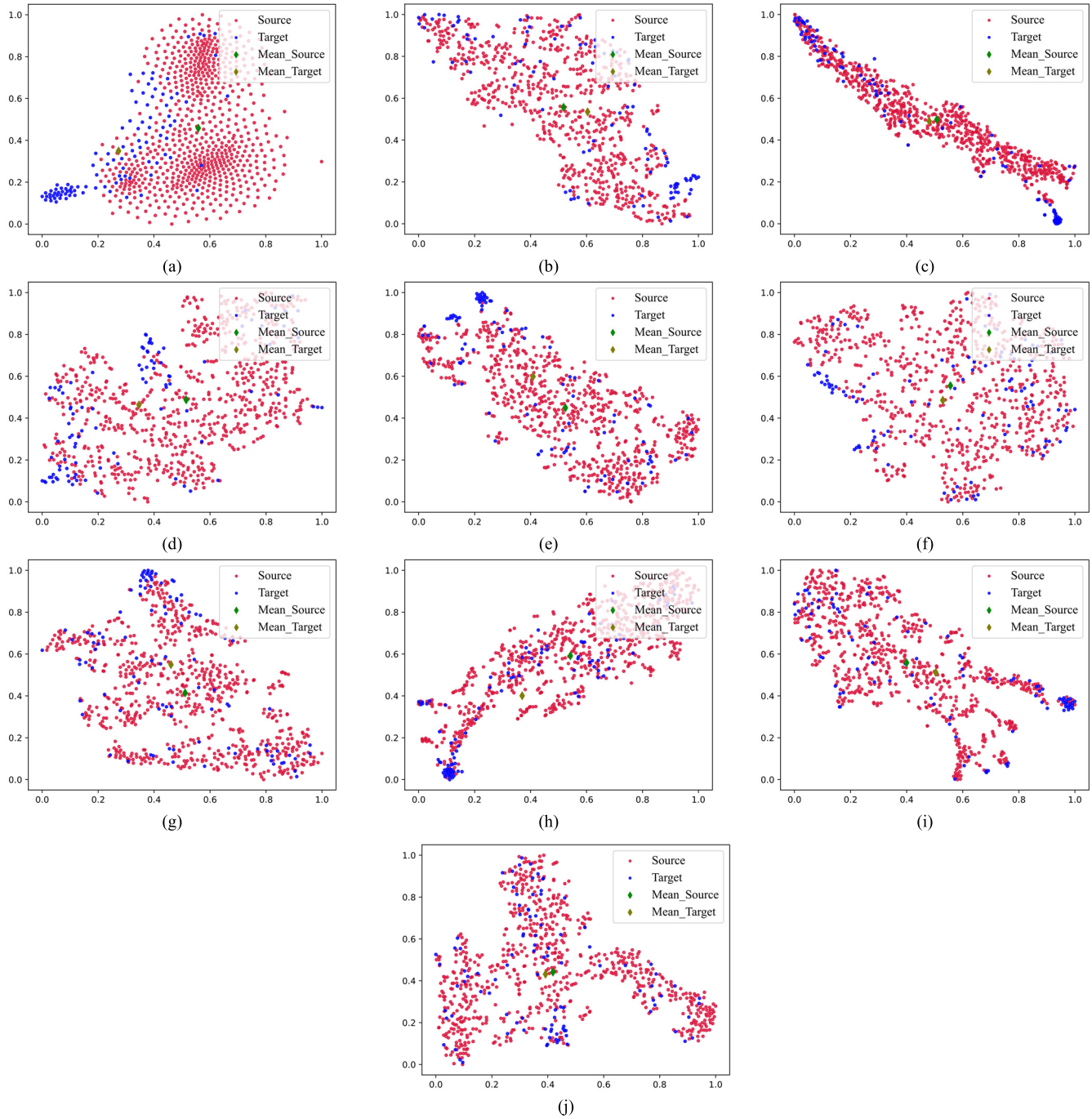


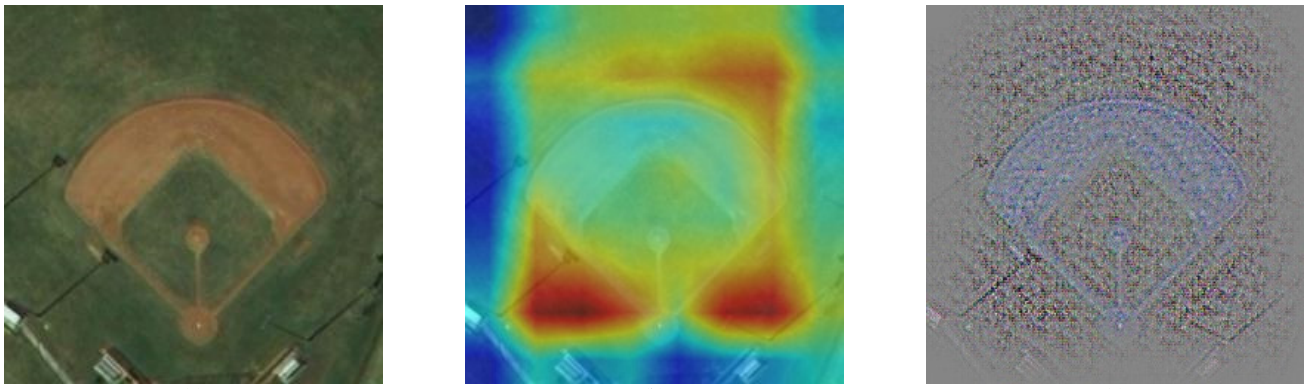
Fig. 6. Visualization: visualized results of t-SNE dimensionality reduction on the source and target domains on the dense residential class of the N→U task. (a) Original samples. (b) DAAN. (c) DeepCORAL. (d) BNM. (e) CDAN. (f) AMRAN. (g) CDAN+E. (h) DSAN. (i) DeepMEDA. (j) ADA-DDA.

Through further analysis, it is found that, when the number of samples is small, the classification accuracy of the target domain decreases more. For example, on U→A and U→N tasks, the classification accuracy of all the methods does not exceed 80%.

2) Discriminativeness and Semantic Consistency Analysis: To directly reflect the alignment and discriminative capabilities of ADA-DDA, the features extracted by different methods on the N→U task are visualized in Fig. 5(a)–(i) by utilizing t-distributed Stochastic Neighbor Embedding (t-SNE) [67]. It can be found that our proposed method makes the boundaries between classes clearer, and the aggregation between classes is also higher. Although the boundaries of each category of the DeepCORAL method are also clear, the degree of aggregation of each category is not good enough, showing a striped distribution. Compared with AMRAN, the features extracted by ADA-DDA are more discriminative.

To further observe the alignment of each category, we visualize the distribution map of the dense residential category in Fig. 6(a)–(j) on the N→U task. From the visualization

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(a) (b) (c)

Fig. 7. (a) Original image with the semantic category "baseball diamond." (b) Grad-CAM for "baseball diamond." (c) Guided Grad-CAM for "baseball diamond.".

results of the original distribution, both the marginal and conditional distributions of the dense residential category in dataset NWPU and dataset UCM differ significantly. From Fig. 6(b), DAAN makes the category mean centers of the source domain and the target domain farther away, which is one of the reasons for the lower classification accuracy of this method. The comparison algorithms have a small number of samples aggregated and do not make the conditional alignment more uniform. Our proposed method aligns the mean center of the category very well, and there is no phenomenon that individual samples are very clustered.

To more intuitively demonstrate semantic consistency, we use gradient-weighted class activation mapping (Grad-CAM) and Guided Grad-CAM [68] to visualize the basis for the ADA-DDA method to make decisions. Grad-CAM uses the class-specific gradient information flowing into the final convolutional layer of a CNN to produce a coarse localization map of the important regions in the image, which can also reflect whether the high-level semantic information is extracted. Moreover, Guided Grad-CAM can generate a high-resolution class-discriminative visualization. The Grad-CAM and Guided Grad-CAM results for the category "baseball diamond" visualized using the model trained on N→U are shown in Fig. 7. Fig. 7(b) shows that the ADA-DDA method is highly discriminative and semantic consistency, that is, the explanation of the semantic category "baseball diamond" only highlights the areas related to the "baseball diamond." When utilizing Guided Grad-CAM to visualize the semantic category "baseball diamond" in Fig. 7(c), Guided Grad-CAM not only highlights the area of the baseball diamond but also the shape of the baseball diamond's boundary while ignoring a subset of insignificant distracting areas.

From the above analysis, ADA-DDA can not only obtain semantically meaningful domain invariant space to ensure that the object properties are aligned but also ensure semantic consistency.

D. Runtime Analysis

The results of training and testing time in one epoch are listed in Table III. All test codes are in the same hardware environment. The least time-consuming training is DeepCORAL (72.21s), which only uses the ResNet-50 network and coral loss. The most complicated is DeepMEDA, which takes up to 170.02 s. Our proposed ADA-DDA takes 126.02 s, but it has achieved a huge performance improvement that is worth it. In the test, the time consumption of all methods is very close because the network does not need to calculate the backward gradient at this time, as long as the classification result is obtained.

TABLE III
TIME COST OF DIFFERENT METHODS ON TASK N→A

	Training(s)	Test(s)
DAAN	89.54	27.55
DeepCORAL	72.21	25.70
BNM	110.67	27.18
CDAN	94.79	25.78
AMRAN	109.63	30.97
CDAN+E	99.45	27.82
DSAN	158.71	28.25
DeepMEDA	170.02	27.28
ADA-DDA (ours)	126.02	37.07

E. Ablation Study

To demonstrate the effectiveness of different parts of our ADA-DDA, we train all the model variants that have the same experimental parameters as those defined earlier. We test them on the whole transfer tasks. As shown in Table IV, we conduct ablation experiments on four different settings: 1) method that only optimizes AMRAN (A); 2) method that optimizes both AMRAN and ADA structure (A+B); 3) method that optimizes both AMRAN and DDF (A+C); and 4) our method (A+B+C).

From the experimental results in Table V, we obtain the following observations.

1) Due to the small number of samples in the UCM dataset and the large distribution discrepancy between the UCM dataset and the AID dataset, the DDF may not obtain the optimal value. Therefore, on the U→A task, the accuracy of A+C is lower than that of using A alone.

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TABLE IV
SETUP OF ABLATION EXPERIMENTS

	AMRAN	ADAS	DDF
A	✓		
A+B	✓	✓	
A+C	✓		✓
A+B+C (ours)	✓	✓	✓

TABLE V
ACCURACY (%) OF ABLATION EXPERIMENTS ON EACH CROSS-DOMAIN SCENE CLASSIFICATION TASK (RESNET-50 AS BACKBONE)

	U→A	U→N	A→U	A→N	N→U	N→A	AVG
A	74.08	68.09	75.50	86.80	78.50	89.43	78.73
A+B	75.56	67.50	80.25	87.20	85.00	89.54	80.84
A+C	69.43	67.39	82.88	87.71	83.38	90.43	80.20
A+B+C (ours)	**77.78**	**74.76**	**87.50**	**90.70**	**89.63**	**91.98**	**85.39**

However, from the perspective of AVG accuracy, both ADA structure (B) and DDF (C) can greatly improve the performance of AMRAN independently.

2) When ADA structure and DDF are used at the same time, our method improves the accuracy more significantly.
3) Our method is the most superior on all tasks, which reflects that our ADA-DDA extracts discriminative features and well aligns the marginal and conditional distributions.

V. CONCLUSION

In this article, a novel end-to-end domain adaptation approach is proposed for remote sensing cross-domain scene classification, referred to as ADA-DDA. By constructing an ADA structure, the ADA-DDA can align the marginal distribution of the source and target domains, and eliminate the influence of dissimilar overall distribution. It adopts a DDF to balance the relative importance of marginal and conditional distributions.

The proposed method will continue to be improved in the future. The improvement is mainly in the following two aspects:

1) For the problem of fewer source domain samples and poor classification performance, we will reduce the impact of the number of samples on the model performance.
2) Due to the high complexity of the proposed method, the structure of ADA-DDA will be streamlined to improve the efficiency of the algorithm.

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